the one in which the insertion point is located. The macro bases its determination of a page on the current pagination of the document, which is affected by the printer driver you're using and, of course, the formatting characteristics of the document.

MICROSOFT EXCEL XP

Blank And Non-Blank Cells

Figuring out the number of non-blank cells in a range is not entirely straightforward. The COUNTBLANK function returns the number of blank cells in a range, but what if you want to count the number of non-blank cells in the same range? One way is to use the COUNTA function:

=COUNTA(B1:B13)

The problem with this formula is that it doesn't return the complementary value to what COUNTBLANK returns. In other words, the result of COUNTA added to the result of COUNTBLANK doesn't equal the total number of cells in the original range. The reason for this is that both COUNTBLANK and COUNTA treat formulas differently. COUNTBLANK includes, as blank, formulas that return a blank value. COUNTA does not consider such cells blank (even though a blank is returned), so it includes them in its count. If you consider non-blank cells to be those that are not returned by COUNTBLANK, you will need to use this formula:

=(ROWS(B1:B13)*COLUMNS(B1:B13))-COUNTBLANK(B1:B13)

This subtracts the COUNTBLANK result from the total number of cells in the same range.

Counting Unique Values

Excel provides several different ways in which you can count unique values in a long list. Say you have a worksheet in which there is a list of first names, in cells A1:A100, which can contain duplicates. Say you need to determine the number of unique names in the list. First define a named range that represents the list of first names. In the following examples, we assume that the range is named FirstName. If the list contains only text entries and no blank cells, the following will provide a count:

=SUM(1/COUNTIF(FirstName,FirstName))

This should be entered as an array formula, by pressing **[Ctrl]** + **[Shift]** + **[Enter]**. If the list contains blank cells, then the formula becomes a bit more complex. The following long array formula will work if there are blanks:

=SUM(IF(FREQUENCY(IF(LEN(Countries)>0,MATCH(Countries,Countries,0),""), IF(LEN(Countries)>0,MATCH(Countries, Countries,0),""))>0,1))

Assigning Names To Formulas

Once created, formulas are valuable, and you may need to use them over and over again. Unfortunately, Excel has no way to paste commonly-used formulas. But there's something you can do to make your formulas more accessible: assign names to your formulas in the following manner.

Enter your formula. Select the cell containing the formula and press **[F2]**. This brings Excel to Edit mode. Hold down **[Shift]** as you use the cursor control keys to select the entire formula, including the equals sign at its beginning. Press **[Ctrl]** + **[C]**. Then press **[Esc]** to get out of edit mode. The cell with the formula is still selected.

Choose Name from the Insert menu, then choose Define. Excel displays the Define Name dialog box. In the Names in Workbook box, enter the name you want assigned to the formula. Select whatever is in the Refers To box at the bottom of the dialog box, and press **[Ctrl]** + **[V]**. The cell reference is replaced with the formula on the Clipboard.

You need to make sure there are no dollar signs in the formula. If there are, select and delete them. This method of using formulas, therefore, does not work well with absolute references.

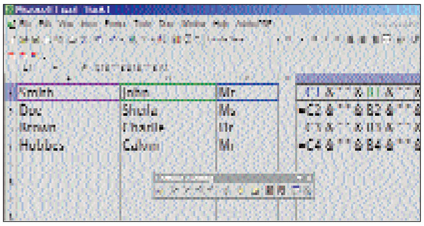
Now, whenever you want to use the formula, you simply enter an equal sign and the name you gave to the formula. Even though the name shows in the cell, the formula assigned to the name is actually used in doing the calculation. Since the formula uses only relative references since you got rid of the dollar signs, it is always relative to where you use the name in the worksheet.

Formulas And Values

Excel normally displays the results of your formulas, but you can also force it to display the formulas themselves. If you're like most people, you choose Tools > Options, then on the View tab, make sure the Formulas checkbox is selected. A much faster way to get the same result is to press **[Ctrl]** + **[`]**. The second key is the one to the left of [1]. The shortcut is a toggle, which means you can press it repeatedly to switch between the display of formulas and results.

Using the TEXT function

Say you want a description for your data. One approach is to put the description near the cell containing the data—for instance, a numeric value could go in cell B1, and the unit description in cell C1.



Use the TEXT function in Excel

Another approach is to put the description text and the numeric value together. Creating text strings does this. So to display "2 + 2 is 4", you'd use

="2 + 2 is " & 2+2

The disadvantage of this approach is that formatting the value takes more effort; since the result is a text string, numeric cell formatting does not apply. For example, considering the above formula, say you need to display two decimal places. You might select Format > Cell > Number tab and then choose a Number format that has two decimal places, but the results would not change. (The result of the formula is text, not a number.)

To affect the value formatting, use the TEXT function. To force the above results to display the value to two decimal places, use the following formula.

="2 + 2 is " & TEXT(2+2, "0.00")

Here's an example that displays "Today is" along with the current date. Enter the following formula:

="Today is " & TEXT(NOW(),"dddd, mmm dd, yyyy")

The quotation marks are important, since you're constructing a text string.